

Bluetooth Networks and Security Protocols

Module 33

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Hello friends I am Professor yogesh Chaba from Department of computer science and engineering of Guru jambheshwar University of Science and Technology Hisar harana I welcome you all to the lecture series of cyber and information security in this lecture we will study about Bluetooth networks and security protocols which is basically used for data transmission on wireless personal area network first of all see the contents which I shall cover after covering brief introduction about Bluetooth technology I shall take up Bluetooth layered protocol architecture which will be followed by frame format of Bluetooth then I shall discuss versions of Bluetooth and i e standards for w pen technology in last I shall take up security architecture of

Bluetooth technology so now let us start Bluetooth is a wireless technology standard for exchanging data over short distances using short wavelength UHF radio waves it operates in the ism band from 2.4 to 2.4856 and mobile devices and building personal area networks in 1994 Ericson company became interested in connecting its mobile devices to other devices without cables together with four other companies IBM Intel Nokia and Toshiba it formed a special interest group they developed a wireless standard for interconnecting computing and communication devices and accessories using short range low power inexpensive Wireless radios this technology got named Bluetooth after the 10th Century King Herald Bluetooth who United disant

Danish tribes into a single Kingdom the idea of this name was proposed in 1997 by by Jim card who developed a system that allowed mobile phones to communicate with computers Bluetooth special interest group SI defined standards for manufacturing Bluetooth devices in July 1999 the Bluetooth SI issued a 1,500 page specification of version 1.0 shortly thereafter the i e standards 3 minutes, 1 second group for wireless personal area network 802.15 adopted the Bluetooth document as a basis and started working on it Bluetooth specification SI is for a complete system that is from the physical layer to the application layer but i e 802.15 defines only only the physical and data link layers the basic unit of a Bluetooth

system is a piconet as shown in figure which consist of a master node and up to seven active slave nodes within a distance of 10 m multiple piconets can exist in the same large room and can even be connected via a bridge node as shown in figure an interconnected collection of piconets is called a scatternet all communication is between the master and the slave direct slave to slave communication is not possible after a brief background let us see layer protocol architecture of Bluetooth the Bluetooth standard has many protocols grouped Loosely into layers the basic Bluetooth protocol architecture is shown in figure the bottom layer is the physical radio layer which corresponds fairly well to the physical layer in the OSI and 802 models it deals with radio

transmission and modulation using ISM band from 2.4 to 5 GHz, 1 second 2485 GHz frequency the baseband layer is somewhat analogous to the mac sub layer but also includes elements of the physical layer it deals with how the master controls time slots and how these slots are grouped into frames the radio layer moves the bits from Master to slave or vice versa this layer turns the raw bit stream into frames and defines some key formats in the simplest form the master in each piconet defines a series of 625 microsec time slots with the Master's transmission starting in the even slot and the slave transmission starting in the odd slots this is traditional time division multiplexing with a master getting half

the slots and slaves sharing the other half each frame is transmitted over a logical Channel called a link between the master and the slave two kind of Link exist the first is the ACL that is asynchronous connectionless link which is used for packet switch data available at irregular intervals the other is the SCO that is synchronous connection oriented link for realtime data such as Telephone Connections this type of channel is allocated a fixed slot in each Direction due to the time critical nature of SCO links frames sent over them are never retransmitted instead forward error correction can be used to provide High reliability next layer link manager handles the establishment of logical

channels between devices including power management authentication and quality of service The Logical link control adaptation protocol often called L2cap Shields the upper layers from the details of transmission the L2cap layer has three major functions first it accepts packets of up to 64 KB from the upper layers and breaks them into frames for transmission at the far end the frames are reassembled into packets again second it handles the multiplexing and demultiplexing of multiple packets sources third L2cap handles the quality of service requirements as the name suggest the audio and control protocols deal with the audio and control respectively the applications can get at them directly without having to go

through the L2 cap protocol the next layer contains a mix of different protocols the RFCOMM radio frequency communication is the protocol that emulates the standard serial Port found on PCs for connecting the keyboard mouse and modem among other devices the telephony protocol is a realtime protocol used for the three speech oriented protocols it also manages call setup and termination finally the service Discovery protocol is used to locate services within the

network the top layer is where the applications and profiles are located they make use of the protocols in lower layers to get their work done so this was layered protocol architecture of Bluetooth now let us see frame format of Bluetooth the frame format of Bluetooth

is shown in figure it begins with an SS code that usually identifies the master so that slaves within radio range of two masters can tell which traffic is for them next comes a 54 bit header containing typical Max sublayer Fields then comes the data field of up to 2744 bits for a five slot transmission for a single time slot the format is same except that the data field is 240 bits let us take a quick look at details of the header the address field identifies which of the eight active devices the frame is intended for the type type field identifies the frame type whether it is ACL SE Pole or null type it also tells the type of error correction used in the data field and how many slots long the frame is the flow bit is for flow control

which is asserted by a slave when its buffer is full and cannot receive any more 11 minutes, 1 second data the acknowledgement bit is used to piggy back an AK onto a frame the sequence bit is used to number the frames to detect retransmissions the protocol is stop and wait so one bit is enough then comes the 8bit header checkm the entire 8bit header is repeated three times to form the 54 bit header shown in figure on the receiving side a simple circuit examines all three copies of each bit if all three are same the bit is accepted if not the majority opinion wins Bluetooth SI has defined versions for Bluetooth and i e has has defined standards for wireless pan let us first see versions defined by Si the versions

were formalized by the Bluetooth special interest group SI the SI was formally announced on 20th May 1998 today it has a membership of over 20,000 companies worldwide it was initially established by Ericson IBM Intel Toshiba and Nokia and later it was joined by many other companies all versions of the Bluetooth standard support downward compatibility let us see each version one by one Bluetooth v1.0 and v1.0b these are basic most Bluetooth which had many problems and manufacturers had difficulty making their products interoperable Bluetooth v1.1 it was ratified as I E standard 802.15 2002 many errors found in the version 1.0b specifications were fixed in this version it added possibility of known and cryp red channels along with

received signal strength indicator Bluetooth v1.2 this version supports faster connection and Discovery protocols it supports adaptive frequency hoping spread Spectrum which improves resistance to radio frequency interference by avoiding the use of crowded frequencies in the hoping sequence it supports higher transmission speeds in practice up to 721 kilobits per second than in v1.1 it was ratified as IA standard 82.1 15.12 2005 Bluetooth v2.0 plus EDR this version of the Bluetooth core specification was released in 2004 the main difference is the introduction of an enhanced data rate EDR for faster data transfer the nominal rate of EDR is about 3 megabits per second although the Practical data transfer rate is 2.1 megabits per second

EDR uses a combination of gfsk and phase shift Key 15 minutes, 1 second modulation Bluetooth 2.1 plus EDR Bluetooth core specification version 2.1 plus EDR was adopted by the Bluetooth SIG on 26 July 2007 the headline feature of v2.1 is secure simple pairing SSP this improves the pairing experience for Bluetooth devices while increasing the use and strength of security Bluetooth v3.0 plus HS this version was adopted by the Bluetooth SIG on 21st April 2009 Bluetooth v3.0 plus HS that is high speed provides theoretical data transfer speed of up to 20 4 megabits per second though not over the Bluetooth Link itself instead the Bluetooth Link is used for negotiation and establishment and the high data rate Traffic is carried over a 802.11

link Bluetooth v4.0 the Bluetooth SIG completed the Bluetooth core specification version 4.0 called bluetooth smart and has been adopted on 30th June 2010 it includes classic Bluetooth Bluetooth highspeed and Bluetooth low energy LE protocols Bluetooth LE previously known as vibri is a subset of Bluetooth v4.0 with an entirely new protocol stack for Rapid buildup of simple links Bluetooth v4.1 the Bluetooth SIG announced formal adoption of the Bluetooth v4.1 specification on 4th December 2013 this specification is an incremental software update to Bluetooth specification v4.0 and not a hardware update features include increased coexistence support for LTE bulk data transfer exchange rates and Aid developer Innovation by allowing devices

to support multiple roles simultaneously Bluetooth v4.2 Bluetooth v4.2 was released on December 2 2014 it introduces some key features such as data length extension that require a hardware update but some older Bluetooth Hardware may receive some Bluetooth v4.2 features such as privacy updates via firmware update also now after Bluetooth version let us see i e standards for wireless personal area network I 802.15 is a working group of The Institute of electrical and electronic Engineers i e which specifies Wireless personal area network standards now we will take up each standard one by one as shown in figure i e 802.1 15.1 W pen Bluetooth this standard is based on Bluetooth technology it defines physical layer phy and media access

control Mac specification for wiress connectivity with fixed portable and moving devices within or entering personal operating space standards were issued in 2002 and 2005 I 82.1 15.2 coexist this standard addresses the coexistence of Wireless personal area networks wpan with other wireless devices operating in unlicensed frequency bands such as Wireless local area networks W Lan the i e 802.15 point2 2003 standard was published in 2003 and task group two went into hibernation i e 802.1 15.3 High rate W pen i e 802.15 3 2003 is a Mac and phy standard for high rate of 11 to 55 Mbps in Wireless personal area networks this standard has three standards i p 802.1 15.3 a 802.1 15.3 B 2006 and 802.1

15.3 C 2009 next is i e 802.15.4 low rate W pen i e 802.15 .4 2003 low rate W pen deals with a low data rate but very long battery life and very low complexity the standard defines both the physical and data link layers of the OSI model this standard has two versions first is W pan low

rate alternative PHY 4 a formally called IEEE 802.15.4 A2 2007 second is revision and enhancement 4B formally called as IEEE 802.15.4 2006 IEEE 802.15.5 Mesh networking it provides the architectural framework enabling W devices to promote 22 minutes, 1 second interoperable stable and scalable Wireless Mesh networking this standard is composed of two parts low rate WPAN meshless and high rate WPAN mesh networks the low rate mesh is built on IEEE 802.15.4

2006 meanwhile the high rate mesh utilizes IEEE 802.15.3 3B MAC next is IEEE 802.15.6 body area networks in December 2011 the IEEE 802.15.6 task group approved a draft of a standard for body area network BAN Technologies focus is on devices in or around the human body to serve a variety of applications including medical consumer electronics and personal entertainment other standards are IEEE 802.15.7 that is visible light communication IEEE 802.15.8 that is peer aware Communications IEEE 802.15.9 that is Key Management protocol and last is IEEE 802.15.10 that is layer 2 routing after versions and standards we now move on to security architecture of Bluetooth three basic security services are

specified in the Bluetooth standard authentication it is for verifying the identity of communicating devices based on their Bluetooth device address Bluetooth does not provide native user authentication second is confidentiality it is for preventing information compromise caused by EES dropping by ensuring that only authorized devices can access and view transmitting data third is authorization it is for allowing the control of resources by ensuring that a device is authorized to use a service before permitting it to do so Bluetooth does not address other security services such as audit integrity and non repudiation if such services are needed they should be provided through additional means the family of Bluetooth BR that is

basic rate EDR that is enhanced data rate HS that is highspeed specifications defines four security modes each Bluetooth device must operate in one of these modes let us see these modes one by one security mode one in this mode devices are considered non-secure security functionality like authentication and encryption is never initiated leaving the device and connections susceptible to attackers in effect Bluetooth devices in this mode are indiscriminate and do not employ any mechanisms to prevent other Bluetooth enabled devices from establishing connections however if a Remote device initiates security such as a pairing authentication or encryption request a security mode one device will participate all v2.0 and earlier devices

can support security mode 1 and v2.1 and later devices can use security mode one for backward compatibility with other devices but NIST recommends never use security mode one for Bluetooth devices second is security mode two in this mode security procedures may be initiated after link establishment but before logical Channel establishment for this security mode a local security manager as specified in the Bluetooth architecture controls access to specific services the centralized security manager maintains policies for Access Control and interfaces with other protocols and device users security mode three this mode is the link

level enforced security mode in which a Bluetooth device initiates security procedures before the physical link is fully

established Bluetooth devices operating in security mode 3 mandate authentication and encryption for all connections to and from the device all v2.0 and earlier devices can support security mode 3 but v2.1 and later devices can only support it for backward compatibility purposes last is security mode 4 similar to security mode 2 security mode 4 introduced in Bluetooth v2.1 plus EDR is a service level and fr security mode in which security procedures are initiated after physical and logical link setup security mode 4 uses secure simple pairing SSP in which elliptic curve Dey Helman key agreement replaces Legacy key agreement for link key generation however the device authentication and encryption algorithm

are identical to the algorithm in Bluetooth v2.0 plus EDR and earlier versions so friends this was Bluetooth networks and security protocols finally summarizing we learned about Bluetooth lead protocol architecture which was followed by frame format of Bluetooth then we discussed versions and it standards for Bluetooth technology in last we covered security architecture of Bluetooth technology hope the concepts explained in this lecture were understandable and helpful hope to see you in next lecture till then goodbye enjoy the day thank you